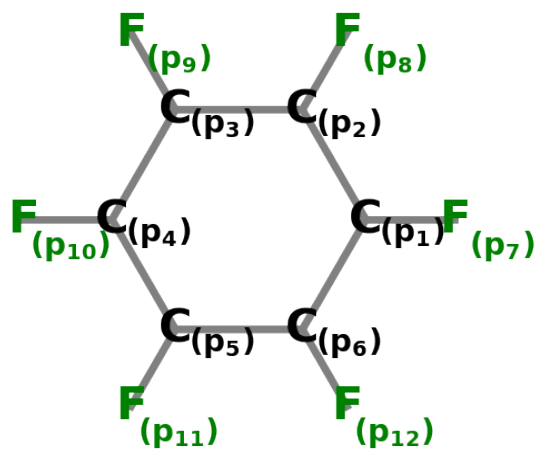


Hexafluorobenzene



Point Group: **D_{6h}**

p orbitals

Symmetry behavior of the p-orbitals:

$$\Gamma_{red} = 12E ; -4C'2 ; -12\sigma h ; 4\sigma v$$

$$\Gamma_{irred} = 2B2g \oplus 2E1g \oplus 2A2u \oplus 2E2u$$

- SALC of B2g: $\frac{\sqrt{6}(p_1 - p_2 + p_3 - p_4 + p_5 - p_6)}{6}$
- SALC of B2g: $\frac{\sqrt{6}(-p_{10} + p_{11} - p_{12} + p_7 - p_8 + p_9)}{6}$
- SALC of E1g: $\frac{\sqrt{3}(2p_1 + p_2 - p_3 - 2p_4 - p_5 + p_6)}{6}$
- SALC of E1g: $\frac{\sqrt{3}(p_1 + 2p_2 + p_3 - p_4 - 2p_5 - p_6)}{6}$
- SALC of E1g: $\frac{\sqrt{3}(-p_1 + p_2 + 2p_3 + p_4 - p_5 - 2p_6)}{6}$
- SALC of E1g: $\frac{\sqrt{3}(-p_{10} - 2p_{11} - p_{12} + p_7 + 2p_8 + p_9)}{6}$
- SALC of A2u: $\frac{\sqrt{6}(p_1 + p_2 + p_3 + p_4 + p_5 + p_6)}{6}$
- SALC of A2u: $\frac{\sqrt{6}(p_{10} + p_{11} + p_{12} + p_7 + p_8 + p_9)}{6}$
- SALC of E2u: $\frac{\sqrt{3}(2p_1 - p_2 - p_3 + 2p_4 - p_5 - p_6)}{6}$
- SALC of E2u: $\frac{\sqrt{3}(-p_1 + 2p_2 - p_3 - p_4 + 2p_5 - p_6)}{6}$
- SALC of E2u: $\frac{\sqrt{3}(-p_1 - p_2 + 2p_3 - p_4 - p_5 + 2p_6)}{6}$
- SALC of E2u: $\frac{\sqrt{3}(p_{10} + p_{11} - 2p_{12} + p_7 + p_8 - 2p_9)}{6}$

Put together these SALCs form the following Hamilton matrices:

$$H_{B2g} = \begin{pmatrix} \tilde{\alpha} - 2\tilde{\beta} & 0 \\ 0 & \tilde{\alpha}_F \end{pmatrix}$$

$$H_{E1g} = \begin{pmatrix} \tilde{\alpha} + \tilde{\beta} & \frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & -\frac{\tilde{\alpha}}{2} - \frac{\tilde{\beta}}{2} & 0 \\ \frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & \tilde{\alpha} + \tilde{\beta} & \frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & 0 \\ -\frac{\tilde{\alpha}}{2} - \frac{\tilde{\beta}}{2} & \frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & \tilde{\alpha} + \tilde{\beta} & 0 \\ 0 & 0 & 0 & \tilde{\alpha}_F \end{pmatrix}$$

$$H_{A2u} = \begin{pmatrix} \tilde{\alpha} + 2\tilde{\beta} & 6\tilde{\beta}_F \\ 6\tilde{\beta}_F & \tilde{\alpha}_F \end{pmatrix}$$

$$H_{E2u} = \begin{pmatrix} \tilde{\alpha} - \tilde{\beta} & -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & 0 \\ -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & \tilde{\alpha} - \tilde{\beta} & -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & 0 \\ -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & -\frac{\tilde{\alpha}}{2} + \frac{\tilde{\beta}}{2} & \tilde{\alpha} - \tilde{\beta} & 0 \\ 0 & 0 & 0 & \tilde{\alpha}_F \end{pmatrix}$$

Solving Hückels secular equations, that follow from these H, leads to:

$$\begin{aligned} & \bullet \text{ orbital of A2u with energy } = \frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_F}{2} + \tilde{\beta} - \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}}{2} \\ & \left(- \frac{12\tilde{\beta}_F}{\sqrt{144 \left| \frac{\tilde{\beta}_F}{\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}} \right|^2} + 1 \left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2} \right) \sqrt{144}} \right. \\ & \quad \left. \sqrt{6} \left(-12\tilde{\beta}_F (\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4 + \mathbf{p}_5 + \mathbf{p}_6) + \left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2} \right) \right) \right. \\ & \quad \left. = \frac{6\sqrt{144 \left| \frac{\tilde{\beta}_F}{\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}} \right|^2} + 1 \left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} + \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2} \right) \sqrt{144}} \right. \\ & \quad \left. \right) \end{aligned}$$

$$\begin{aligned} & \bullet \text{ orbital of A2u with energy } = \frac{\tilde{\alpha}}{2} + \frac{\tilde{\alpha}_F}{2} + \tilde{\beta} + \frac{\sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}}{2} \\ & \left(- \frac{12\tilde{\beta}_F}{\sqrt{144 \left| \frac{\tilde{\beta}_F}{\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}} \right|^2} + 1 \left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2} \right) \sqrt{144}} \right. \\ & \quad \left. \sqrt{6} \left(-12\tilde{\beta}_F (\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4 + \mathbf{p}_5 + \mathbf{p}_6) + \left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2} \right) \right) \right. \\ & \quad \left. = \frac{\sqrt{6} \left(-12\tilde{\beta}_F (\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4 + \mathbf{p}_5 + \mathbf{p}_6) + \left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2} \right) \right)}{6\sqrt{144 \left| \frac{\tilde{\beta}_F}{\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2}} \right|^2} + 1 \left(\tilde{\alpha} - \tilde{\alpha}_F + 2\tilde{\beta} - \sqrt{\tilde{\alpha}^2 - 2\tilde{\alpha}\tilde{\alpha}_F + 4\tilde{\alpha}\tilde{\beta} + \tilde{\alpha}_F^2 - 4\tilde{\alpha}_F\tilde{\beta} + 4\tilde{\beta}^2 + 144\tilde{\beta}_F^2} \right) \sqrt{144}} \right. \\ & \quad \left. \right) \end{aligned}$$

$$\begin{aligned} & \text{orbital of } E_{1g} \text{ with energy} = \frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2} \\ & \left(\begin{array}{cccc} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \end{array} \right) * \begin{pmatrix} \text{1. SALC in } E_{1g} \\ \text{2. SALC in } E_{1g} \\ \text{3. SALC in } E_{1g} \\ \text{4. SALC in } E_{1g} \end{pmatrix} = \left(\begin{array}{cccc} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \end{array} \right) * \begin{pmatrix} \frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} + \\ \frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} + \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{3} - \\ - \frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} + \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} - \\ - \frac{\sqrt{3}p_{10}}{6} - \frac{\sqrt{3}p_{11}}{3} - \frac{\sqrt{3}p_{12}}{6} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{3} \end{pmatrix} \\ & = \frac{\sqrt{6}(p_1 + p_2 - p_4 - p_5)}{4} \end{aligned}$$

Error in Orthogonalization of SALCs

$$\begin{aligned} & \text{orbital of } E_{1g} \text{ with energy} = \frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2} \\ & \left(-\frac{\sqrt{2}}{2} \quad 0 \quad \frac{\sqrt{2}}{2} \quad 0 \right) * \begin{pmatrix} \text{1. SALC in } E_{1g} \\ \text{2. SALC in } E_{1g} \\ \text{3. SALC in } E_{1g} \\ \text{4. SALC in } E_{1g} \end{pmatrix} = \left(-\frac{\sqrt{2}}{2} \quad 0 \quad \frac{\sqrt{2}}{2} \quad 0 \right) * \begin{pmatrix} \frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} \\ \frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} + \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{3} \\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} + \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{3} \\ -\frac{\sqrt{3}p_{10}}{6} - \frac{\sqrt{3}p_{11}}{3} - \frac{\sqrt{3}p_{12}}{6} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{6} \end{pmatrix} \\ & = \frac{\sqrt{6}(-p_1 + p_3 + p_4 - p_6)}{4} \end{aligned}$$

Error in Orthogonalization of SALCs

- orbital of E1g with energy = 0
(

$$\begin{pmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in E1g} \\ \text{2. SALC in E1g} \\ \text{3. SALC in E1g} \\ \text{4. SALC in E1g} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{3} - \\ \frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{3} + \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} - \\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} + \frac{\sqrt{3}p_3}{3} + \frac{\sqrt{3}p_4}{6} - \\ -\frac{\sqrt{3}p_{10}}{6} - \frac{\sqrt{3}p_{11}}{3} - \frac{\sqrt{3}p_{12}}{6} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{3} + \frac{\sqrt{3}p_9}{6} \end{pmatrix}$$

$$= 0$$

Error in Orthogonalization of SALCsError in Orthogonalization of SALCsError in Orthogonalization of SALCs

- orbital of E1g with energy = $\tilde{\alpha}_F$
(

$$\begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in E1g} \\ \text{2. SALC in E1g} \\ \text{3. SALC in E1g} \\ \text{4. SALC in E1g} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} + \frac{\sqrt{3}p_6}{6} \\ \frac{\sqrt{3}p_1}{3} + \frac{\sqrt{3}p_2}{3} + \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{3} - \frac{\sqrt{3}p_6}{6} \\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} + \frac{\sqrt{3}p_3}{3} + \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} - \frac{\sqrt{3}p_6}{3} \\ -\frac{\sqrt{3}p_{10}}{6} - \frac{\sqrt{3}p_{11}}{3} - \frac{\sqrt{3}p_{12}}{6} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{3} + \frac{\sqrt{3}p_9}{6} \end{pmatrix}$$

$$= \frac{\sqrt{3}(-p_{10} - 2p_{11} - p_{12} + p_7 + 2p_8 + p_9)}{6}$$

Error in Orthogonalization of SALCs

- orbital of B2g with energy = $\tilde{\alpha}_F$
(

$$\begin{pmatrix} 0 & 1 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in B2g} \\ \text{2. SALC in B2g} \end{pmatrix} = \begin{pmatrix} 0 & 1 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{6}p_1}{6} - \frac{\sqrt{6}p_2}{6} + \frac{\sqrt{6}p_3}{6} - \frac{\sqrt{6}p_4}{6} + \frac{\sqrt{6}p_5}{6} - \frac{\sqrt{6}p_6}{6} \\ -\frac{\sqrt{6}p_{10}}{6} + \frac{\sqrt{6}p_{11}}{6} - \frac{\sqrt{6}p_{12}}{6} + \frac{\sqrt{6}p_7}{6} - \frac{\sqrt{6}p_8}{6} + \frac{\sqrt{6}p_9}{6} \end{pmatrix}$$

$$= \frac{\sqrt{6}(-p_{10} + p_{11} - p_{12} + p_7 - p_8 + p_9)}{6}$$

- orbital of B2g with energy = $\tilde{\alpha} - 2\tilde{\beta}$
(

$$\begin{pmatrix} 1 & 0 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in B2g} \\ \text{2. SALC in B2g} \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{6}p_1}{6} - \frac{\sqrt{6}p_2}{6} + \frac{\sqrt{6}p_3}{6} - \frac{\sqrt{6}p_4}{6} + \frac{\sqrt{6}p_5}{6} - \frac{\sqrt{6}p_6}{6} \\ -\frac{\sqrt{6}p_{10}}{6} + \frac{\sqrt{6}p_{11}}{6} - \frac{\sqrt{6}p_{12}}{6} + \frac{\sqrt{6}p_7}{6} - \frac{\sqrt{6}p_8}{6} + \frac{\sqrt{6}p_9}{6} \end{pmatrix}$$

$$= \frac{\sqrt{6}(p_1 - p_2 + p_3 - p_4 + p_5 - p_6)}{6}$$

- orbital of E2u with energy = 0

$$\left(\begin{array}{cccc} \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0 \end{array} \right) * \left(\begin{array}{l} 1. \text{ SALC in E2u} \\ 2. \text{ SALC in E2u} \\ 3. \text{ SALC in E2u} \\ 4. \text{ SALC in E2u} \end{array} \right) = \left(\begin{array}{cccc} \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0 \end{array} \right) * \left(\begin{array}{l} \frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} \\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} + \frac{\sqrt{3}p_5}{3} \\ -\frac{\sqrt{3}p_1}{6} - \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} \\ \frac{\sqrt{3}p_{10}}{6} + \frac{\sqrt{3}p_{11}}{6} - \frac{\sqrt{3}p_{12}}{3} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{6} \end{array} \right) = 0$$

Error in Orthogonalization of SALCs

- orbital of E2u with energy = $\tilde{\alpha}_F$

$$\left(\begin{array}{cccc} 0 & 0 & 0 & 1 \end{array} \right) * \left(\begin{array}{l} 1. \text{ SALC in E2u} \\ 2. \text{ SALC in E2u} \\ 3. \text{ SALC in E2u} \\ 4. \text{ SALC in E2u} \end{array} \right) = \left(\begin{array}{cccc} 0 & 0 & 0 & 1 \end{array} \right) * \left(\begin{array}{l} \frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} - \frac{\sqrt{3}p_6}{6} \\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} + \frac{\sqrt{3}p_5}{3} - \frac{\sqrt{3}p_6}{6} \\ -\frac{\sqrt{3}p_1}{6} - \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} + \frac{\sqrt{3}p_6}{3} \\ \frac{\sqrt{3}p_{10}}{6} + \frac{\sqrt{3}p_{11}}{6} - \frac{\sqrt{3}p_{12}}{3} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{6} - \frac{\sqrt{3}p_9}{3} \end{array} \right) = \frac{\sqrt{3}(p_{10} + p_{11} - 2p_{12} + p_7 + p_8 - 2p_9)}{6}$$

Error in Orthogonalization of SALCs

- orbital of E2u with energy = $\frac{3\tilde{\alpha}}{2} - \frac{3\tilde{\beta}}{2}$

$$\left(\begin{array}{cccc} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \end{array} \right) * \left(\begin{array}{l} 1. \text{ SALC in E2u} \\ 2. \text{ SALC in E2u} \\ 3. \text{ SALC in E2u} \\ 4. \text{ SALC in E2u} \end{array} \right) = \left(\begin{array}{cccc} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \end{array} \right) * \left(\begin{array}{l} \frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} \\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} + \frac{\sqrt{3}p_5}{3} \\ -\frac{\sqrt{3}p_1}{6} - \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} \\ \frac{\sqrt{3}p_{10}}{6} + \frac{\sqrt{3}p_{11}}{6} - \frac{\sqrt{3}p_{12}}{3} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{6} \end{array} \right) = \frac{\sqrt{6}(-p_1 + p_2 - p_4 + p_5)}{4}$$

Error in Orthogonalization of SALCs

- orbital of E2u with energy = $\frac{3\tilde{\alpha}}{2} - \frac{3\tilde{\beta}}{2}$

$$\left(\begin{array}{cccc} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0 \end{array} \right) * \left(\begin{array}{l} 1. \text{ SALC in E2u} \\ 2. \text{ SALC in E2u} \\ 3. \text{ SALC in E2u} \\ 4. \text{ SALC in E2u} \end{array} \right) = \left(\begin{array}{cccc} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0 \end{array} \right) * \left(\begin{array}{l} \frac{\sqrt{3}p_1}{3} - \frac{\sqrt{3}p_2}{6} - \frac{\sqrt{3}p_3}{6} + \frac{\sqrt{3}p_4}{3} - \frac{\sqrt{3}p_5}{6} \\ -\frac{\sqrt{3}p_1}{6} + \frac{\sqrt{3}p_2}{3} - \frac{\sqrt{3}p_3}{6} - \frac{\sqrt{3}p_4}{6} + \frac{\sqrt{3}p_5}{3} \\ -\frac{\sqrt{3}p_1}{6} - \frac{\sqrt{3}p_2}{6} + \frac{\sqrt{3}p_3}{3} - \frac{\sqrt{3}p_4}{6} - \frac{\sqrt{3}p_5}{6} \\ \frac{\sqrt{3}p_{10}}{6} + \frac{\sqrt{3}p_{11}}{6} - \frac{\sqrt{3}p_{12}}{3} + \frac{\sqrt{3}p_7}{6} + \frac{\sqrt{3}p_8}{6} \end{array} \right) = \frac{\sqrt{6}(-p_1 + p_3 - p_4 + p_6)}{4}$$

Error in Orthogonalization of SALCs

s orbitals

Symmetry behavior of the s-orbitals:

$$\Gamma_{red} = 12E ; 4C''2 ; 12\sigma h ; 4\sigma v$$

$$\Gamma_{irred} = 2A1g \oplus 2E2g \oplus 2B1u \oplus 2E1u$$

- SALC of A1g: $\frac{\sqrt{6}(s_1+s_2+s_3+s_4+s_5+s_6)}{6}$
- SALC of A1g: $\frac{\sqrt{6}(s_{10}+s_{11}+s_{12}+s_7+s_8+s_9)}{6}$
- SALC of E2g: $\frac{\sqrt{3}(2s_1-s_2-s_3+2s_4-s_5-s_6)}{6}$
- SALC of E2g: $\frac{\sqrt{3}(-s_1+2s_2-s_3-s_4+2s_5-s_6)}{6}$
- SALC of E2g: $\frac{\sqrt{3}(-s_1-s_2+2s_3-s_4-s_5+2s_6)}{6}$
- SALC of E2g: $\frac{\sqrt{3}(s_{10}+s_{11}-2s_{12}+s_7+s_8-2s_9)}{6}$
- SALC of B1u: $\frac{\sqrt{6}(s_1-s_2+s_3-s_4+s_5-s_6)}{6}$
- SALC of B1u: $\frac{\sqrt{6}(-s_{10}+s_{11}-s_{12}+s_7-s_8+s_9)}{6}$
- SALC of E1u: $\frac{\sqrt{3}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{6}$
- SALC of E1u: $\frac{\sqrt{3}(s_1+2s_2+s_3-s_4-2s_5-s_6)}{6}$
- SALC of E1u: $\frac{\sqrt{3}(-s_1+s_2+2s_3+s_4-s_5-2s_6)}{6}$
- SALC of E1u: $\frac{\sqrt{3}(-s_{10}-2s_{11}-s_{12}+s_7+2s_8+s_9)}{6}$

Put together these SALCs form the following Hamilton matrices:

$$H_{A1g} = \begin{pmatrix} \tilde{\alpha}_s + 2\tilde{\beta}_s & 6\beta_{s_F} \\ 6\beta_{s_F} & \alpha_{s_F} \end{pmatrix}$$

$$H_{E2g} = \begin{pmatrix} \tilde{\alpha}_s - \tilde{\beta}_s & -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & 0 \\ -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & \tilde{\alpha}_s - \tilde{\beta}_s & -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & 0 \\ -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & -\frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & \tilde{\alpha}_s - \tilde{\beta}_s & 0 \\ 0 & 0 & 0 & \alpha_{s_F} \end{pmatrix}$$

$$H_{B1u} = \begin{pmatrix} \tilde{\alpha}_s - 2\tilde{\beta}_s & 0 \\ 0 & \alpha_{s_F} \end{pmatrix}$$

$$H_{E1u} = \begin{pmatrix} \tilde{\alpha}_s + \tilde{\beta}_s & \frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & -\frac{\tilde{\alpha}_s}{2} - \frac{\tilde{\beta}_s}{2} & 0 \\ \frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & \tilde{\alpha}_s + \tilde{\beta}_s & \frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & 0 \\ -\frac{\tilde{\alpha}_s}{2} - \frac{\tilde{\beta}_s}{2} & \frac{\tilde{\alpha}_s}{2} + \frac{\tilde{\beta}_s}{2} & \tilde{\alpha}_s + \tilde{\beta}_s & 0 \\ 0 & 0 & 0 & \alpha_{s_F} \end{pmatrix}$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of A1g with energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_F}}{2} + \tilde{\beta}_s - \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2}}{2}$

$$\left(- \frac{12\beta_{s_F}}{\sqrt{144 \left| \frac{\beta_{s_F}}{\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2}} \right|^2} + 1 \left(\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2} \right)} \right. \\
= \frac{\sqrt{6} \left(-12\beta_{s_F} (s_1 + s_2 + s_3 + s_4 + s_5 + s_6) + \left(\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2} \right)} \right)}{6\sqrt{144 \left| \frac{\beta_{s_F}}{\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2}} \right|^2} + 1 \left(\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s + \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2} \right)}$$

- orbital of A1g with energy = $\frac{\tilde{\alpha}_s}{2} + \frac{\alpha_{s_F}}{2} + \tilde{\beta}_s + \frac{\sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2}}{2}$

$$\left(- \frac{12\beta_{s_F}}{\sqrt{144 \left| \frac{\beta_{s_F}}{\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2}} \right|^2} + 1 \left(\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2} \right)} \right. \\
= \frac{\sqrt{6} \left(-12\beta_{s_F} (s_1 + s_2 + s_3 + s_4 + s_5 + s_6) + \left(\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2} \right)} \right)}{6\sqrt{144 \left| \frac{\beta_{s_F}}{\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2}} \right|^2} + 1 \left(\tilde{\alpha}_s - \alpha_{s_F} + 2\tilde{\beta}_s - \sqrt{\tilde{\alpha}_s^2 - 2\tilde{\alpha}_s\alpha_{s_F} + 4\tilde{\alpha}_s\tilde{\beta}_s + \alpha_{s_F}^2 - 4\alpha_{s_F}\tilde{\beta}_s + 4\tilde{\beta}_s^2 + 144\beta_{s_F}^2} \right)}$$

- orbital of E1u with energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$

$$\left(\frac{\sqrt{2}}{2} \quad \frac{\sqrt{2}}{2} \quad 0 \quad 0 \right) * \begin{pmatrix} 1. \text{ SALC in E1u} \\ 2. \text{ SALC in E1u} \\ 3. \text{ SALC in E1u} \\ 4. \text{ SALC in E1u} \end{pmatrix} = \left(\frac{\sqrt{2}}{2} \quad \frac{\sqrt{2}}{2} \quad 0 \quad 0 \right) * \begin{pmatrix} \frac{\sqrt{3}s_1}{3} + \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} + \\ \frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} + \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{3} - \\ - \frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} + \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} - \\ - \frac{\sqrt{3}s_{10}}{6} - \frac{\sqrt{3}s_{11}}{3} - \frac{\sqrt{3}s_{12}}{6} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{3} \end{pmatrix} \\
= \frac{\sqrt{6} (s_1 + s_2 - s_4 - s_5)}{4}$$

Error in Orthogonalization of SALCs

- orbital of E1u with energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$

$$\left(-\frac{\sqrt{2}}{2} \quad 0 \quad \frac{\sqrt{2}}{2} \quad 0 \right) * \begin{pmatrix} 1. \text{ SALC in E1u} \\ 2. \text{ SALC in E1u} \\ 3. \text{ SALC in E1u} \\ 4. \text{ SALC in E1u} \end{pmatrix} = \left(-\frac{\sqrt{2}}{2} \quad 0 \quad \frac{\sqrt{2}}{2} \quad 0 \right) * \begin{pmatrix} \frac{\sqrt{3}s_1}{3} + \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} + \\ \frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} + \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{3} - \\ - \frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} + \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} - \\ - \frac{\sqrt{3}s_{10}}{6} - \frac{\sqrt{3}s_{11}}{3} - \frac{\sqrt{3}s_{12}}{6} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{3} \end{pmatrix} \\
= \frac{\sqrt{6} (-s_1 + s_3 + s_4 - s_6)}{4}$$

Error in Orthogonalization of SALCs

- orbital of E1u with energy = 0

$$\begin{pmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in E1u} \\ \text{2. SALC in E1u} \\ \text{3. SALC in E1u} \\ \text{4. SALC in E1u} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{3} & -\frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{3}s_1}{3} + \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} + \frac{\sqrt{3}s_6}{6} \\ \frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} + \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{3} - \frac{\sqrt{3}s_6}{6} \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} + \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} - \frac{\sqrt{3}s_6}{3} \\ -\frac{\sqrt{3}s_{10}}{6} - \frac{\sqrt{3}s_{11}}{3} - \frac{\sqrt{3}s_{12}}{6} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{3} + \frac{\sqrt{3}s_9}{6} \end{pmatrix} \\
 = 0
 \end{pmatrix}$$

Error in Orthogonalization of SALCsError in Orthogonalization of SALCsError in Orthogonalization of SALCs

- orbital of E1u with energy = α_{s_F}

$$\begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in E1u} \\ \text{2. SALC in E1u} \\ \text{3. SALC in E1u} \\ \text{4. SALC in E1u} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{3}s_1}{3} + \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} + \frac{\sqrt{3}s_6}{6} \\ \frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} + \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{3} - \frac{\sqrt{3}s_6}{6} \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} + \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} - \frac{\sqrt{3}s_6}{3} \\ -\frac{\sqrt{3}s_{10}}{6} - \frac{\sqrt{3}s_{11}}{3} - \frac{\sqrt{3}s_{12}}{6} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{3} + \frac{\sqrt{3}s_9}{6} \end{pmatrix} \\
 = \frac{\sqrt{3}(-s_{10} - 2s_{11} - s_{12} + s_7 + 2s_8 + s_9)}{6}
 \end{pmatrix}$$

Error in Orthogonalization of SALCs

- orbital of E2g with energy = 0

$$\begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in E2g} \\ \text{2. SALC in E2g} \\ \text{3. SALC in E2g} \\ \text{4. SALC in E2g} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{3}s_1}{3} - \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} + \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} + \frac{\sqrt{3}s_6}{6} \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} + \frac{\sqrt{3}s_5}{3} - \frac{\sqrt{3}s_6}{6} \\ -\frac{\sqrt{3}s_1}{6} - \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} + \frac{\sqrt{3}s_6}{3} \\ \frac{\sqrt{3}s_{10}}{6} + \frac{\sqrt{3}s_{11}}{6} - \frac{\sqrt{3}s_{12}}{3} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{6} - \frac{\sqrt{3}s_9}{3} \end{pmatrix} \\
 = 0
 \end{pmatrix}$$

Error in Orthogonalization of SALCsError in Orthogonalization of SALCsError in Orthogonalization of SALCs

- orbital of E2g with energy = α_{s_F}

$$\begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} \text{1. SALC in E2g} \\ \text{2. SALC in E2g} \\ \text{3. SALC in E2g} \\ \text{4. SALC in E2g} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{3}s_1}{3} - \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} + \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} + \frac{\sqrt{3}s_6}{6} \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} + \frac{\sqrt{3}s_5}{3} - \frac{\sqrt{3}s_6}{6} \\ -\frac{\sqrt{3}s_1}{6} - \frac{\sqrt{3}s_2}{6} + \frac{\sqrt{3}s_3}{3} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{6} + \frac{\sqrt{3}s_6}{3} \\ \frac{\sqrt{3}s_{10}}{6} + \frac{\sqrt{3}s_{11}}{6} - \frac{\sqrt{3}s_{12}}{3} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{6} - \frac{\sqrt{3}s_9}{3} \end{pmatrix} \\
 = \frac{\sqrt{3}(s_{10} + s_{11} - 2s_{12} + s_7 + s_8 - 2s_9)}{6}
 \end{pmatrix}$$

Error in Orthogonalization of SALCs

- orbital of E2g with energy = $\frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}_s}{2}$
(

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in E2g} \\ 2. \text{ SALC in E2g} \\ 3. \text{ SALC in E2g} \\ 4. \text{ SALC in E2g} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{3}s_1}{3} - \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} + \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} + \frac{\sqrt{3}s_5}{3} \\ -\frac{\sqrt{3}s_1}{6} - \frac{\sqrt{3}s_2}{3} + \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{3} \\ \frac{\sqrt{3}s_{10}}{6} + \frac{\sqrt{3}s_{11}}{6} - \frac{\sqrt{3}s_{12}}{3} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{6} \end{pmatrix}$$

$$= \frac{\sqrt{6}(-s_1 + s_2 - s_4 + s_5)}{4}$$

Error in Orthogonalization of SALCs

- orbital of E2g with energy = $\frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}_s}{2}$
(

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in E2g} \\ 2. \text{ SALC in E2g} \\ 3. \text{ SALC in E2g} \\ 4. \text{ SALC in E2g} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{3}s_1}{3} - \frac{\sqrt{3}s_2}{6} - \frac{\sqrt{3}s_3}{6} + \frac{\sqrt{3}s_4}{3} - \frac{\sqrt{3}s_5}{6} \\ -\frac{\sqrt{3}s_1}{6} + \frac{\sqrt{3}s_2}{3} - \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} + \frac{\sqrt{3}s_5}{3} \\ -\frac{\sqrt{3}s_1}{6} - \frac{\sqrt{3}s_2}{3} + \frac{\sqrt{3}s_3}{6} - \frac{\sqrt{3}s_4}{6} - \frac{\sqrt{3}s_5}{3} \\ \frac{\sqrt{3}s_{10}}{6} + \frac{\sqrt{3}s_{11}}{6} - \frac{\sqrt{3}s_{12}}{3} + \frac{\sqrt{3}s_7}{6} + \frac{\sqrt{3}s_8}{6} \end{pmatrix}$$

$$= \frac{\sqrt{6}(-s_1 + s_3 - s_4 + s_6)}{4}$$

Error in Orthogonalization of SALCs

- orbital of B1u with energy = α_{s_F}
(

$$\begin{pmatrix} 0 & 1 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in B1u} \\ 2. \text{ SALC in B1u} \end{pmatrix} = \begin{pmatrix} 0 & 1 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{6}s_1}{6} - \frac{\sqrt{6}s_2}{6} + \frac{\sqrt{6}s_3}{6} - \frac{\sqrt{6}s_4}{6} + \frac{\sqrt{6}s_5}{6} - \frac{\sqrt{6}s_6}{6} \\ -\frac{\sqrt{6}s_{10}}{6} + \frac{\sqrt{6}s_{11}}{6} - \frac{\sqrt{6}s_{12}}{6} + \frac{\sqrt{6}s_7}{6} - \frac{\sqrt{6}s_8}{6} + \frac{\sqrt{6}s_9}{6} \end{pmatrix}$$

$$= \frac{\sqrt{6}(-s_{10} + s_{11} - s_{12} + s_7 - s_8 + s_9)}{6}$$

- orbital of B1u with energy = $\tilde{\alpha}_s - 2\tilde{\beta}_s$
(

$$\begin{pmatrix} 1 & 0 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in B1u} \\ 2. \text{ SALC in B1u} \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{6}s_1}{6} - \frac{\sqrt{6}s_2}{6} + \frac{\sqrt{6}s_3}{6} - \frac{\sqrt{6}s_4}{6} + \frac{\sqrt{6}s_5}{6} - \frac{\sqrt{6}s_6}{6} \\ -\frac{\sqrt{6}s_{10}}{6} + \frac{\sqrt{6}s_{11}}{6} - \frac{\sqrt{6}s_{12}}{6} + \frac{\sqrt{6}s_7}{6} - \frac{\sqrt{6}s_8}{6} + \frac{\sqrt{6}s_9}{6} \end{pmatrix}$$

$$= \frac{\sqrt{6}(s_1 - s_2 + s_3 - s_4 + s_5 - s_6)}{6}$$

Transitions from p orbitals into s orbitals

State with Occupation (2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2) (0 unpaired electrons, mult 1.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $16\tilde{\alpha} + 4\tilde{\alpha}_F$
- Energy of State after Excitation: $\frac{29\tilde{\alpha}}{2} + 4\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0, 2, 2, 2, 2, 2, 2), (0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2p_1-p_2+p_3+2p_4+p_5-p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2s_1-s_2+s_3+2s_4+s_5-s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $16\tilde{\alpha} + 4\tilde{\alpha}_F$
- Energy of State after Excitation: $\frac{29\tilde{\alpha}}{2} + 4\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 1, 1, 1, 1, 0, 2, 2, 2, 2, 2, 2) (4 unpaired electrons, mult 5.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 1, 1, 1, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 1, 1, 1, 0, 2, 2, 2, 2, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$

- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $13\tilde{\alpha} + 5\tilde{\alpha}_F - 3\tilde{\beta}$
- Energy of State after Excitation: $\frac{23\tilde{\alpha}}{2} + 5\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - \frac{9\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 1, 1, 1, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 0, 1, 1, 0, 2, 2, 2, 2, 2, 2), (0, 0, 1, 0, 0, 0,$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2p_1-p_2+p_3+2p_4+p_5-p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2s_1-s_2+s_3+2s_4+s_5-s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $13\tilde{\alpha} + 5\tilde{\alpha}_F - 3\tilde{\beta}$
- Energy of State after Excitation: $\frac{23\tilde{\alpha}}{2} + 5\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - \frac{9\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2), (0, 1, 0, 0, 0, 0,$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 4\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 4\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2), (0, 0, 1, 0, 0, 0,$

- ψ_3 : linear combination = $\frac{\sqrt{6}(-2p_1-p_2+p_3+2p_4+p_5-p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2s_1-s_2+s_3+2s_4+s_5-s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 4\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 4\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 5\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 5\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2), (0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2p_1-p_2+p_3+2p_4+p_5-p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2s_1-s_2+s_3+2s_4+s_5-s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 5\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 5\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 4\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 4\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 0, 1, 0, 2, 2, 2, 2, 2, 2), (0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2p_1-p_2+p_3+2p_4+p_5-p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2s_1-s_2+s_3+2s_4+s_5-s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 4\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 4\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

State with Occupation (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2), (0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $\frac{\sqrt{6}(2p_1+p_2-p_3-2p_4-p_5+p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_2 : linear combination = $\frac{\sqrt{6}(2s_1+s_2-s_3-2s_4-s_5+s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$

- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 5\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 5\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2), (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 1, 0, 0, 2, 2, 2, 2, 2, 2), (0, 0, 1, 0, 0,$
- ψ_3 : linear combination = $\frac{\sqrt{6}(-2p_1-p_2+p_3+2p_4+p_5-p_6)}{3}$, energy = $\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\beta}}{2}$
- η_3 : linear combination = $\frac{\sqrt{6}(-2s_1-s_2+s_3+2s_4+s_5-s_6)}{3}$, energy = $\frac{3\tilde{\alpha}_s}{2} + \frac{3\tilde{\beta}_s}{2}$
- Δ Energy: $-\frac{3\tilde{\alpha}}{2} + \frac{3\tilde{\alpha}_s}{2} - \frac{3\tilde{\beta}}{2} + \frac{3\tilde{\beta}_s}{2}$
- Energy of State before Excitation: $\frac{29\tilde{\alpha}}{2} + 5\tilde{\alpha}_F - \frac{3\tilde{\beta}}{2}$
- Energy of State after Excitation: $13\tilde{\alpha} + 5\tilde{\alpha}_F + \frac{3\tilde{\alpha}_s}{2} - 3\tilde{\beta} + \frac{3\tilde{\beta}_s}{2}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\frac{2\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_F}{3} + \frac{2\tilde{\alpha}_s}{3} + \frac{\alpha_{s_F}}{3}$
- dipole transition moment: $\frac{3\tilde{\delta}}{2}$

- AND

Dispersion energies following from the transitions

1. State: A = [2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2] $\leftrightarrow$ B = (2, 1, 1, 1, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{54\tilde{\delta}^4}{r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

2. State: A = (2, 1, 1, 1, 1, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = [2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2]

$$E_{dispersion} = -\frac{54\tilde{\delta}^4}{r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

3. State: A = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

4. State: A = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

5. State: A = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

6. State: A = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

7. State: A = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

8. State: A = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

9. State: A = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

10. State: A = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

11. State: A = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

12. State: A = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

13. State: A = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

14. State: A = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

15. State: A = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

16. State: A = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 1, 2, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

17. State: A = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 0, 1, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

18. State: A = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2) $\leftrightarrow$ B = (2, 2, 1, 1, 0, 0, 2, 2, 2, 2, 2, 2)

$$E_{dispersion} = -\frac{243\tilde{\delta}^4}{4r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

Ground-State:

A=[2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2] $\leftrightarrow$ B=[2, 2, 2, 0, 0, 0, 2, 2, 2, 2, 2, 2]

$$E_{dispersion} = -\frac{27\tilde{\delta}^4}{r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$